AI&ML Lab: 24B15CS222

MINI PROJECT

TOPIC : QuestBreak

AI Anti-Doomscrolling App using Behaviour Prediction & Sentimental Analysis

BATCH : G1

SUBMITTED TO :

DR.ANIL KUMAR MAHTO
DR.TARUN AGRAWAL
DR.AMIT MISHRA
DR.SONAL SAURABH

SUBMITTED BY :

1.ARJUN BANSAL - 2401200002
2. NAISHA SHARMA - 2401200016
3. SHWETANK TIWARI - 2401200025
4.AARUSH BAHUGUNA - 2401200026

ABSTRACT

The widespread use of short-form digital platforms such as Instagram and YouTube has led to a behavioral pattern known as *doomscrolling*, where users continuously consume content with minimal cognitive engagement. This habit negatively affects productivity, attention span, and mental well-being.

This project proposes an AI-based gamified web application that transforms passive scrolling into an interactive treasure hunt experience. Instead of endlessly browsing content, users participate in quests that involve solving logical puzzles, riddles, and real-world observation challenges. The application dynamically generates missions based on the user's interests and behavioral patterns.

User activity data is processed using Python libraries such as NumPy and Pandas to analyze engagement patterns and personalize quests. The system also includes leaderboards, progress tracking, and gamification elements to maintain motivation and encourage healthy digital habits.

Machine learning models classify user preferences and recommend personalized tasks, while computer vision techniques validate image-based challenges such as identifying colored objects or specific items in the user's environment.

By combining Artificial Intelligence, Machine Learning, Deep Learning, and Gamification, the system redirects digital dopamine loops into problem-solving activities and real-world interaction, promoting mindful technology usage.

TOPICS OF AI/ML USED

Feature of Proposed System	Course Outcome	Mapping Explanation
Treasure Hunt Puzzle System	CO1 – Python Programming Fundamentals	Implemented using Python constructs such as lists, dictionaries, loops, functions, and conditional statements to manage riddles, puzzles, and quest flow.
User Data Processing	CO1 – Python Libraries	User activity data is stored and processed using NumPy arrays and Pandas DataFrames to analyze engagement patterns and behavior trends.
Quest Recommendation Engine	CO2 – Search Techniques	AI agents generate quests using search strategies such as A*, Hill Climbing, and Best-First Search to determine optimal puzzle difficulty and sequence.
User Interest Prediction	CO3 – Machine Learning	Supervised learning models including Decision Trees, Logistic Regression, Random Forest, and SVM analyze user activity patterns and recommend personalized quests.
User Behaviour Clustering	CO3 – Unsupervised Learning	k-means clustering groups users into categories such as puzzle solvers, explorers, or observation-based players to improve recommendations.
Leaderboard Ranking System	CO3 – Data Analysis and Model Evaluation	Engagement metrics such as quest completion rate, accuracy, and streak duration are analyzed to generate leaderboard rankings.
Progress Tracking System	CO1 / CO3	Python data structures and machine learning models track user progress, calculate performance metrics, and dynamically adjust quest difficulty

Image Verification for Tasks	CO4 – Deep Neural Networks	Convolutional Neural Networks verify uploaded images for tasks such as identifying objects or colors
Visual Recognition Models	CO4 – CNN Architectures	CNN architectures such as LeNet, AlexNet, and VGGNet are implemented using TensorFlow and Keras to classify uploaded images.
Model Visualization	CO4 – Neural Network Implementation	Training accuracy, loss curves, and model performance are visualized using Matplotlib.

CLASS DIAGRAM

Class Name	What It Does	Main Methods
DataPipeline	Loads the simulated data, cleans it, creates engineered features, scales everything, and splits it into training and test sets	simulate_data(), clean_and_engineer(), split_and_scale()
SearchEngine	Contains all four search algorithms (BFS, DFS, A*, Hill Climbing) and the quest bank. Given the user's current state, it finds and returns the best quest	bfs(), dfs(), astar(), hill_climb()
MLClassifier	Trains all six machine learning models and the K-Means clustering. Evaluates and compares them. Provides the craving probability for any new user state	train_all(), predict_craving(), evaluate_all(), feature_importance()
CNNVerifier	Builds and trains LeNet, AlexNet-style, and ResNet50. Takes a photo submitted by the user and returns whether the quest was genuinely completed	build_lenet(), build_alexnet(), build_resnet(), verify_photo(), plot_curves()
QuestBreakApp	The main controller. Connects all other classes and runs the full pipeline: detect craving → generate quest → verify photo → update dopamine score	run(), get_profile(), fire_quest(), update_dopamine()